BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 053 · Arithmetic, Comparison and Casting

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does assignment do to a variable?

Q2. What is the CPU's main job?
A. Execute instructions
B. Keep permanent file backups
C. Display every image
D. Supply electrical power

Q3. A folder contains 30 images of 200 KB each. Using decimal units, give its total size in KB and MB.

Q4. A photographer travels with large files and needs fast transfers. “The cheapest storage is always best.” Correct this. What else matters?

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I Do · Read new knowledge

Key vocabulary

Arithmetic operators — Operators such as +, -, * and / allow programs to calculate results.

Comparison operators — Symbols such as ==, < and >= compare values and produce Boolean results.

Casting — int() converts appropriate text or numeric values to integer so arithmetic can use them.

Order of operations — Parentheses can make intended calculation order clear and avoid ambiguous results.

Algorithm — An ordered set of instructions to solve a problem.

Trace — Following instructions and recording changes to values or output.

New learning

What is Arithmetic operators?: Arithmetic operators — Operators such as +, -, * and / allow programs to calculate results.

In this example: A train fare calculator uses comparison operators; Symbols such as ==, < and >= compare values and produce Boolean results.

Follow the effect: int() converts appropriate text or numeric values to integer so arithmetic can use them. Explain how this changes the eventual program result.

Check understanding: Order of operations is used for a specific reason: Parentheses can make intended calculation order clear and avoid ambiguous results.

Worked example

A train fare calculator is being written. Explain how arithmetic operators and comparison operators are used to make the program work. [4 marks]

1. We are designing a train fare calculator. Identify which data and decisions are essential.

2. Arithmetic operators [1] — Operators such as +, -, * and / allow programs to calculate results. [1].

3. Comparison operators [1] — Symbols such as ==, < and >= compare values and produce Boolean results. [1].

4. Two linked points for each technique give four mark points.

We Do · Practise with guidance

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Guided 1

In a museum ticket price, explain how arithmetic operators helps the program. [2 marks]

Worked method: Arithmetic operators [1] — Operators such as +, -, * and / allow programs to calculate results. [1].

Guided 2

For a retail discount, explain why comparison operators matters. [2 marks]

Worked method: Comparison operators [1] — Symbols such as ==, < and >= compare values and produce Boolean results. [1].

Guided 3

When designing a sports points, explain how casting and order of operations should be used. [4 marks]

Worked method: Casting [1] — int() converts appropriate text or numeric values to integer so arithmetic can use them. [1]. Order of operations [1] — Parentheses can make intended calculation order clear and avoid ambiguous results. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Convert the text "4" into an integer and print how many bits are in four bytes.

```
bytesText = "4"
bytes = ...
print(bytes * 8)
```

write

Question 1

State two programming terms or techniques relevant to this lesson. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 2

A school trip cost needs to work correctly. Explain how comparison operators and casting would help. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 3

A pupil is programming a takeaway bill. Explain the roles of arithmetic operators, casting and order of operations in their solution. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

Operators such as +, -, * and / allow programs to calculate results.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It stores a value in the named variable, replacing its previous value if one existed.

Q2. A — Execute instructions

Q3. 6,000 KB = 6 MB.

Q4. Check capacity, speed, cost and how it will be used. An SSD can suit fast transfers and travel, if its capacity and price meet the photographer's needs.

I Do / We Do pages · Guided practice

Q1. Arithmetic operators [1] — Operators such as +, -, * and / allow programs to calculate results. [1].

Q2. Comparison operators [1] — Symbols such as ==, < and >= compare values and produce Boolean results. [1].

Q3. Casting [1] — int() converts appropriate text or numeric values to integer so arithmetic can use them. [1]. Order of operations [1] — Parentheses can make intended calculation order clear and avoid ambiguous results. [1].

You Do pages · Code First

```
bytesText = "4"
bytes = int(bytesText)
print(bytes * 8)
```

Expected result: 32

You Do pages · Independent questions

Q1. Arithmetic operators [1]; Casting [1].

Q2. Comparison operators [1] — Symbols such as ==, < and >= compare values and produce Boolean results. [1]. Casting [1] — int() converts appropriate text or numeric values to integer so arithmetic can use them. [1].

Q3. Arithmetic operators [1] — Operators such as +, -, * and / allow programs to calculate results. [1]. Casting [1] — int() converts appropriate text or numeric values

to integer so arithmetic can use them. [1]. Order of operations [1] — Parentheses can make intended calculation order clear and avoid ambiguous results. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: